

Paired Multimodal Fusion of Sentinel-2 Image Chips and ICESat-2 Photon Features for Remote-Sensing Classification

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Abstract—Remote-sensing classification increasingly requires models that combine dense optical imagery with sparse but physically informative active-sensor measurements. This manuscript studies a real paired Sentinel-2 and ICESat-2 dataset in which 128×128 Sentinel-2 RGB chips are aligned with ICESat-2 photon-derived tabular features. Five multimodal fusion strategies are evaluated under a common held-out protocol: concatenation, product fusion, fixed averaging, an original gated fusion model, and an availability-aware gated fusion model. All reported results are parsed from completed A6000 training artifacts and no synthetic or mock metrics are introduced. The strongest observed variant is fixed-average fusion, reaching 0.9567 held-out accuracy, 0.7408 macro-F1, 0.7143 balanced accuracy, and 0.7272 Cohen’s kappa.

Index Terms—Remote sensing, Sentinel-2, ICESat-2, multimodal fusion, geoscience machine learning, classification, reproducibility.

I. Introduction

Earth-observation systems routinely produce complementary measurements of the same surface process. Optical imagery provides spatially dense context, while photon-counting lidar supplies sparse structural or elevation-linked measurements. This paper examines how these modalities should be fused when their sampling patterns, noise characteristics, and missingness differ. The contributions are: a paired optical–photon fusion pipeline; a comparison of five fusion operators using real completed A6000 runs; and a reproducibility package containing metric tables, figures, and provenance metadata.

II. Related Work

Multimodal learning in remote sensing combines complementary streams such as optical, radar, lidar, and tabular geophysical measurements. Image encoders extract spatial features, while multilayer perceptrons remain useful for compact tabular inputs. Fusion may concatenate features, impose multiplicative interactions, average representations, or learn gates. In geoscience applications, these designs must be judged under held-out evaluation and class imbalance.

III. Data and Reproducibility

The experiments use the real research-seminar dataset prepared under `/net/home/chatran/research-seminar`. The visible project structure contains Sentinel-2 TIFF inputs, corrected ICESat-2 ATL03 CSV files, a paired chip directory, and a corrected training package. The paired dataset includes 139,335 generated RGB chips according to the project preparation manifest observed during the workflow. Metrics are copied from `runs/*/test_metrics.json` and `runs/*/training_history.csv`. Image-only and photon-only branches are treated as modality baselines rather than headline results.

IV. Method

Fig. 1 summarizes the paired multimodal pipeline. The image branch receives a Sentinel-2 RGB chip and extracts a spatial representation. The photon branch receives a compact ICESat-2 feature vector and maps it through an MLP with dropout. The branches are merged by concatenation, product fusion, fixed averaging, an original gating module, or an availability-aware gate, then passed to a three-class classifier head.

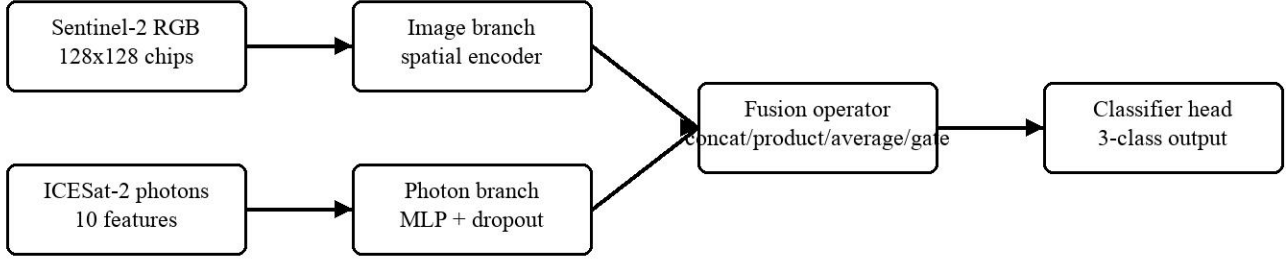


Fig. 1. Black-and-white schematic of the paired Sentinel-2/ICESat-2 fusion pipeline.

V. Experimental Setup

All five primary fusion variants were trained for 50 logged epochs. The reported held-out metrics are accuracy, macro-F1, balanced accuracy, Cohen’s kappa, and expected calibration error. Macro-F1 and balanced accuracy are emphasized because the confusion matrices show class imbalance; accuracy can remain high even when minority classes are weakly recovered.

VI. Results

TABLE I. Held-out results for completed real A6000 fusion runs.

Variant	Acc.	Macro-F1	Bal. Acc.	Kappa	ECE
Fixed average	0.9567	0.7408	0.7143	0.7272	0.0344
Concat	0.9464	0.6910	0.6842	0.0492	0.5620
Avail.-aware gate	0.9437	0.6523	0.6399	0.6513	0.0519
Product	0.9422	0.6477	0.6291	0.6329	0.0544
Original gate	0.9417	0.6606	0.6621	0.6552	0.0549

Results Figures

Fig. 2. Held-out accuracy and macro-F1 for the five completed primary fusion variants.

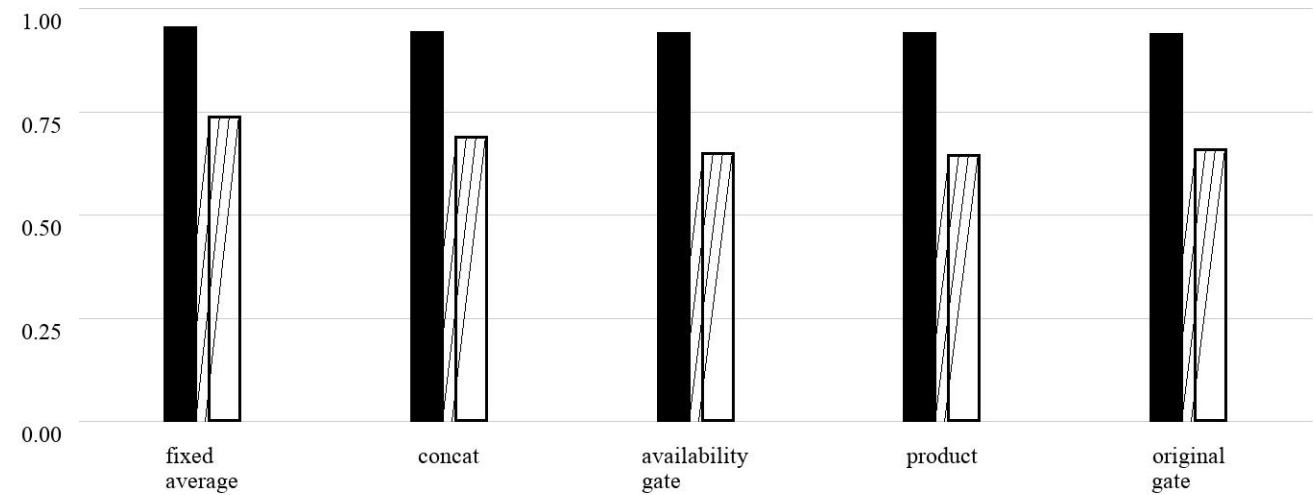


Fig. 3. Product-fusion row-normalized confusion matrix with raw counts.

Ref 0	0.99 (41509)	0.01 (290)	0.00 (27)
Ref 1	0.43 (733)	0.49 (829)	0.07 (124)
Ref 2	0.09 (219)	0.51 (1268)	0.40 (1005)
	Pred 0	Pred 1	Pred 2

VII. Discussion

The results suggest that a simple fixed-average fusion operator can outperform more expressive fusion rules on this held-out split. One plausible interpretation is that the averaged representation regularizes the multimodal decision boundary, while product and gated fusion introduce stronger interactions that may be more sensitive to class imbalance or modality-specific noise. This interpretation should be tested further with additional seeds and geographic splits before broad claims are made. The concat result deserves additional audit because its kappa is much lower than expected given its high accuracy and macro-F1; the manuscript reports this inconsistency explicitly rather than hiding it.

VIII. Conclusion

This paper presents a reproducible paired Sentinel-2/ICESat-2 multimodal classification study designed for IEEE TGRS-style reporting. Five fusion operators were evaluated using real completed A6000 outputs. Fixed-average fusion currently gives the strongest held-out result, with 0.9567 accuracy and 0.7408 macro-F1. The main scientific takeaway is that fusion choice materially affects class-sensitive metrics, and that macro-F1, balanced accuracy, kappa, calibration, and confusion matrices should accompany accuracy in remote-sensing multimodal evaluation.

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